DO NOW — RECALL

Lesson 090 · Inside City Hall: ALU, Control Unit and Regi

Name: _____ Class: _____ Date: _____

1. What does CU stand for?

2. What is the ALU?

3. What does ACC typically store?

4. A full adder has A=1,B=1 and carry-in=0. What are Sum and carry-out?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Control Unit.

2. Model answer

Arithmetic Logic Unit.

3. Model answer

Intermediate arithmetic/logic result.

4. Model answer

$1+1+0=10_2$, so Sum0 and carry-out1.

By the end, you can…

• Control Unit generates control signals and interprets the fetched instruction. It is not the component that performs arithmetic itself.

• ALU completes arithmetic and logic such as addition and comparisons; the ACC may hold a result in an accumulator-style model.

• Registers serve distinct roles: PC next instruction, MAR memory address, MDR transferred contents, CIR current instruction, ACC intermediate calculation.

Vocabulary — know these terms

Control Unit — Directs instruction execution.

ALU — Performs arithmetic and logic.

ACC — Intermediate calculation register.

PC — Address of next instruction.

NEW LEARNING

Read, annotate and explain

Observe

Control Unit generates control signals and interprets the fetched instruction. It is not the component that performs arithmetic itself.

Explain

ALU completes arithmetic and logic such as addition and comparisons; the ACC may hold a result in an accumulator-style model.

Evaluate

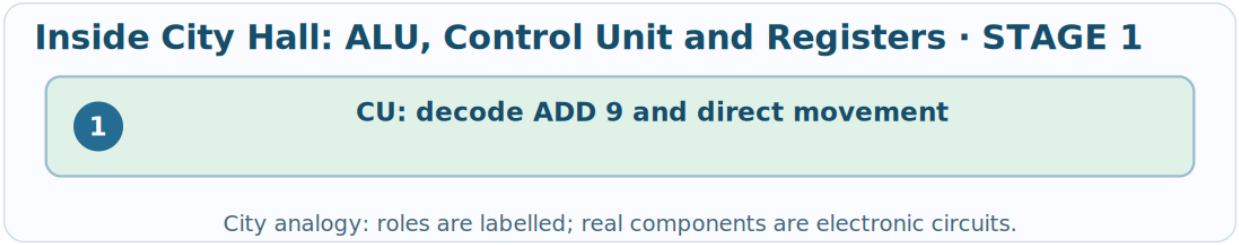
Registers serve distinct roles: PC next instruction, MAR memory address, MDR transferred contents, CIR current instruction, ACC intermediate calculation.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

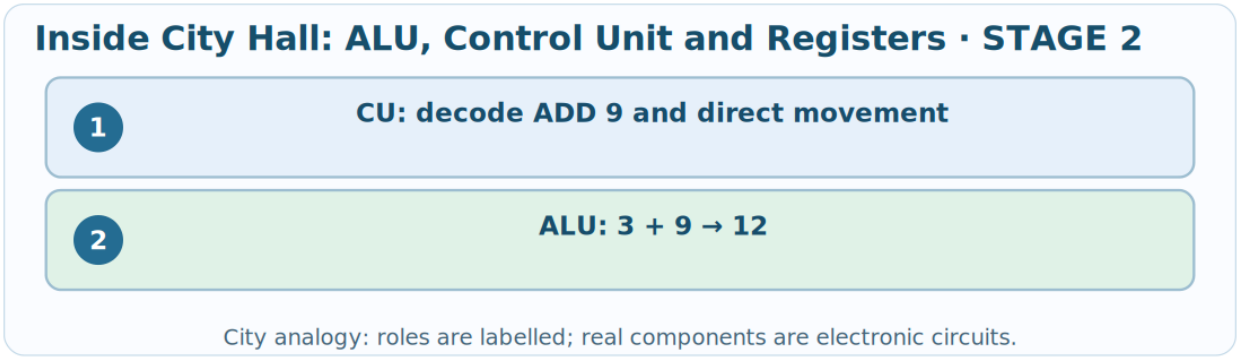
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Inside City Hall: ALU, Control Unit and Registers · STAGE 1; CU: decode ADD 9 and direct movement

New Learning

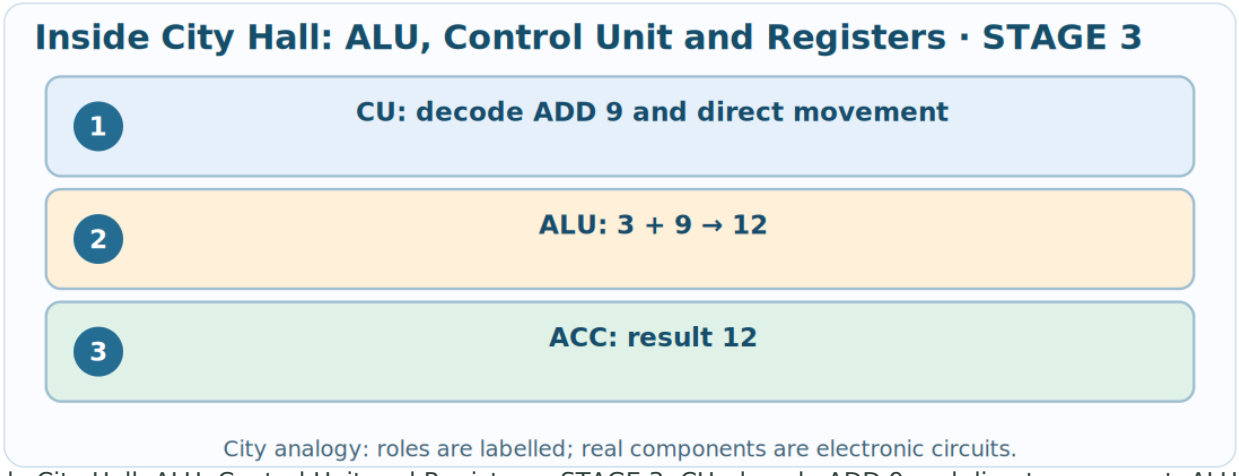


Inside City Hall: ALU, Control Unit and Registers · STAGE 2; CU: decode ADD 9 and direct movement; ALU: 3 + 9 → 12

New Learning · original lesson diagrams

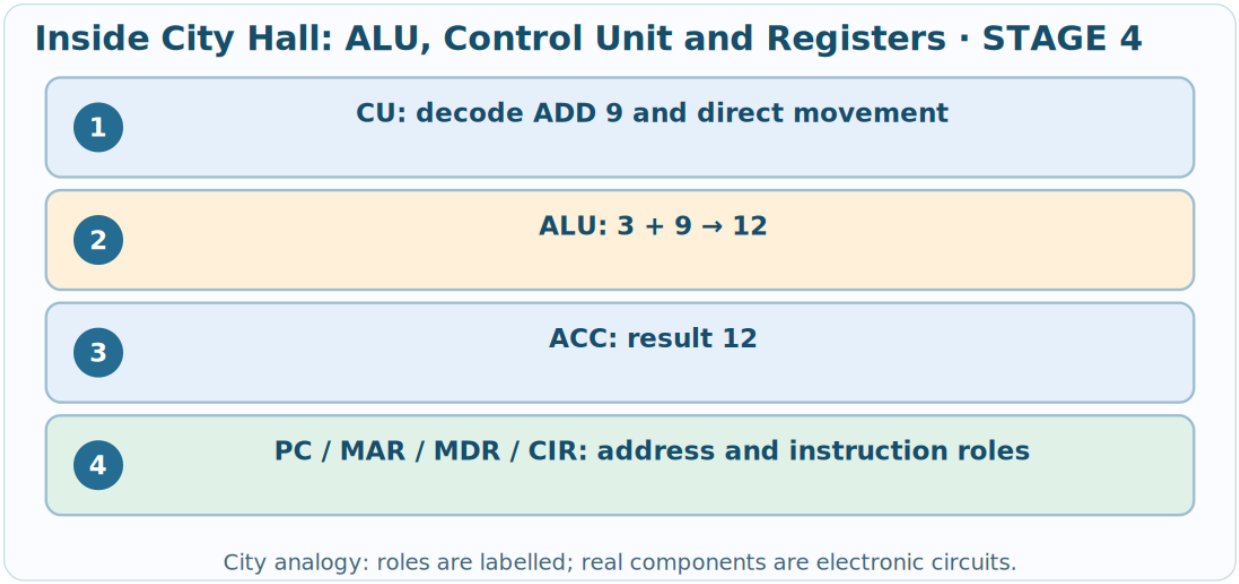
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Inside City Hall: ALU, Control Unit and Registers · STAGE 3; CU: decode ADD 9 and direct movement; ALU: 3 + 9 → 12; ACC: result 12

New Learning



Inside City Hall: ALU, Control Unit and Registers · STAGE 4; CU: decode ADD 9 and direct movement; ALU: 3 + 9 → 12; ACC: result 12; PC / MAR / MDR / CIR: address and instruction roles

I DO — WORKED EXAMPLE

Follow the worked method

ACC starts5; toy instruction ADD 7 is executed. Trace ALU and ACC.

Model response

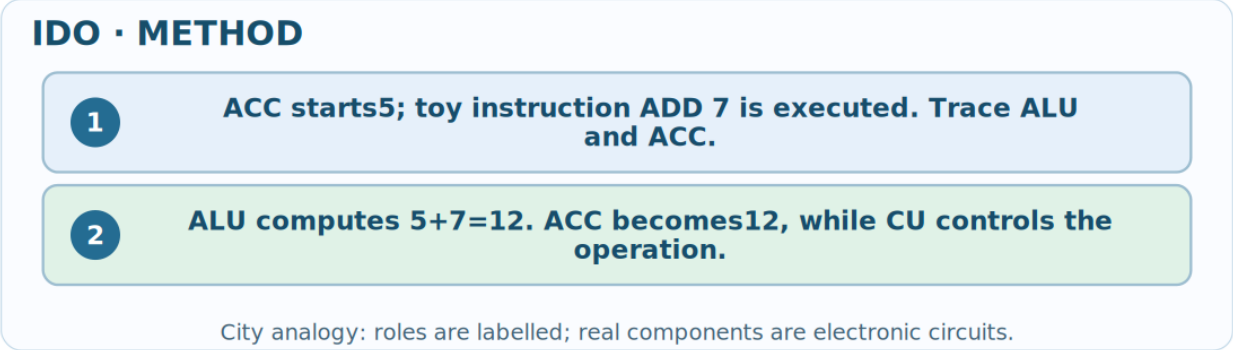
ACC holds intermediate result, not memory address.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

ACC starts5; toy instruction ADD 7 is executed. Trace ALU and ACC.

ACC holds intermediate result, not memory address.



IDO · METHOD; ACC starts5; toy instruction ADD 7 is executed. Trace ALU and ACC.; ALU computes 5+7=12. ACC becomes12, while CU controls the operation.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

ACC starts 14; toy instruction SUB 6. What changes and why?

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

For ACC=3 and toy ADD9, state ALU calculation, ACC result, CU role. [6]

Question 2

Explain different roles of PC, MAR and CIR in a memory fetch. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Pupil claims CU performs calculations while ALU stores next instruction address. Correct both errors. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

The Program Counter is the ALU register storing calculation results.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

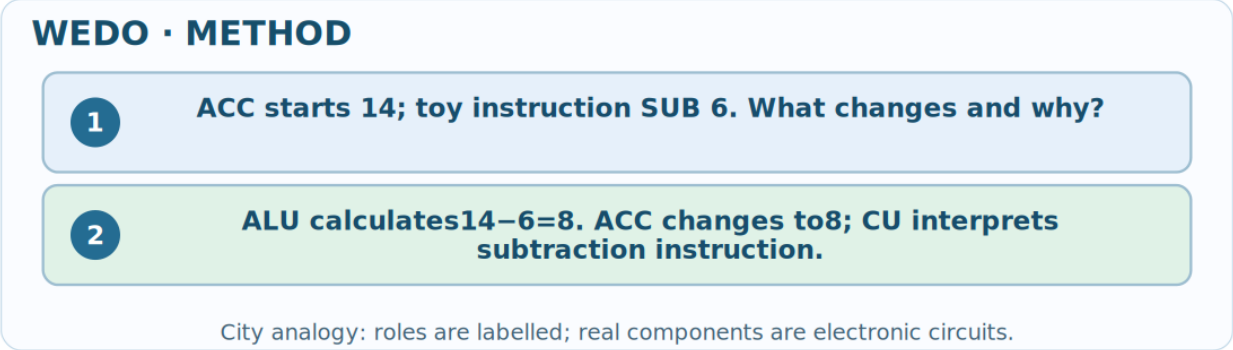
Model response
Do not say PC is overwritten by subtraction; it tracks instruction address.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

ACC starts 14; toy instruction SUB 6. What changes and why?

Do not say PC is overwritten by subtraction; it tracks instruction address.



WEDO · METHOD; ACC starts 14; toy instruction SUB 6. What changes and why?; ALU calculates14−6=8. ACC changes to8; CU interprets subtraction instruction.